Homework 5: Message Passing Neural Networks

CS 1470/2470

Due November 13, 2019 at 11:59pm

1 CS2470-only Questions

1. Our readout layer is essentially taking every node vector, feeding it through a linear layer, summing the result across all nodes, and then returning it as the "representation" of the graph. However, some researchers have criticized global operations like these (e.g taking the max/min/mean across the graph) because your network just spent time learning local relationships through message passing, only for this to be "thrown away" at the global level. In lecture, we have explored some ideas about this ought to be done.

   Consider the following algorithm for a more iterative readout function: After computing the state at each node through rounds of message passing, we cluster these nodes using a clustering algorithm like K-means. Then, we turn every resulting cluster into a single output node by averaging over the state of each node in the cluster.

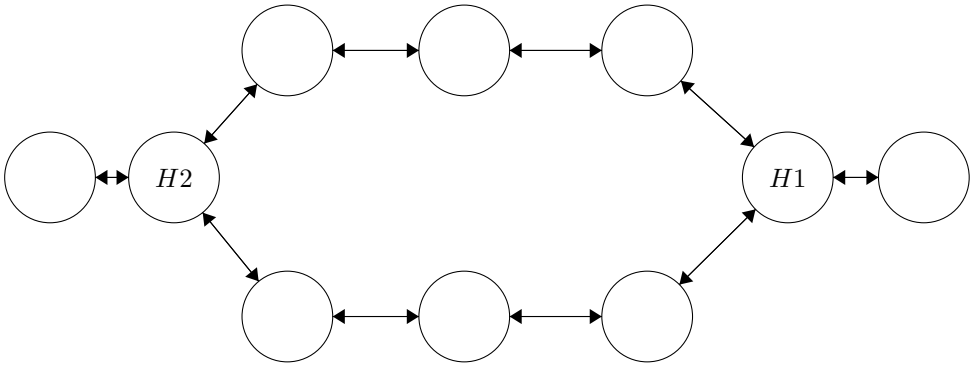
   We posit that this scheme will likely not work. Why is that? And how could you fix this?
   (Hint: You may find this paper helpful in formulating your answer:)

   Answer: K-means does not always guarantee a good pooling results and it may loss some connectivity information from the original nodes. One way to fix this is to use the idea of differentiable pooling implemented in the paper below. This algorithm will generate hierarchical representations of graphs and learns a differentiable soft cluster assignment for nodes at each layer of a deep GNN, mapping nodes to a set of cluster, which then could give a coarse-graining representation of the original network. The network structure simplification can be achieved in such fashion by iterative pooling.

   Ying Z, You J, Morris C, Ren X, Hamilton W, Leskovec J. Hierarchical graph representation learning with differentiable pooling. InAdvances in Neural Information Processing Systems 2018 (pp. 4800-4810).

2. Imagine that you have a graph G with two nodes H_1 and H_2 as follows. If an MPNN is run on this graph and only classifies based on the structure of

the graph (not the node features), will H_1 and H_2 be classified similarly or differently? Is this a disadvantage or an advantage for graph/node classifcation as a whole? Justify your answers.



Answer: If the MPNN only classifies based on the structure of the graph (not the node features), H1 and H2 will be classified similarly since the graph are symmetric. Hence there is no way for the network to identify the difference between two symmetric node H1 and H2 even if they are essentially different. This is a huge disadvantage since chemical molecules could have very same structure or bonds with different atoms on some of the nodes. But the chemical property could be totally different. It could cause a disaster when using such graph without node feature information to identify the type of of such molecule.